

The AirDrop Transport Companion

ECE Team 4

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Abstract

Autonomous drones have the potential to save lives, reduce costs, and provide critical transport during disaster scenarios. In the US, approximately 60,000 people die each year due to traumatic bleeding that could have been prevented with a faster response time. Studies have also shown that drone delivery can reduce medical transport times from 38 to just 14 minutes. In disaster zones where roads are inaccessible, drones can deliver supplies in minutes rather than hours. Beyond these emergency responses, drones also offer economic advantages, reducing delivery, packaging, and operational costs by 40–80%.

Despite this potential, adapting to this is limited because of challenges in reliability and delivery accuracy. Current payload methods that involve winches and parachute drops are susceptible to wind, reducing precision, and can destabilize the drone's flight. To this end, we developed a self-stabilizing payload case to counteract these effects, improving precision and stability for medical and commercial applications. Our payload case can sense its altitude, angle, and if a user is trying to access the package compartment. It responds by adjusting its flight path and can unlock the package compartment by hand wave gestures. In our tests, the payload case stabilizes 58.7% faster with the active control system, and we expect this to improve.

1. Overview

Our project, as described by our sponsor, Dr. Pranav Bhounsule, focuses on a system for a small, unmanned aircraft system or SUAS. It is designed to safely and accurately deliver payloads in scenarios where traditional delivery methods are unreliable or inaccessible. Current payload delivery methods, such as winches and parachute drops, are highly susceptible to wind disturbances, which reduces delivery precision and destabilizes drone flight during operation. This presents a critical challenge for time-sensitive applications such as medical supply delivery and disaster relief, where delivery accuracy and reliability can be a matter of life and death. In disaster zones where roads are inaccessible, or in emergencies requiring rapid medical response, the inability to precisely and reliably deliver a payload significantly limits the effectiveness of drone-based systems.

We will be working alongside the Engineering Design Team (EDT) at UIC to develop the full system. Our senior design team was tasked with solving the stability and precision challenges associated with payload delivery. This includes addressing how a suspended or dropped payload interacts with wind forces, how the drone's flight is affected during delivery, and how payload access can be securely managed upon arrival.

1.1 Project Goals

We aim to develop a system to help improve autonomous drones deliver payloads. Our goal is to create an airdrop control system that would be integrated into drones to safely, quickly, and accurately deliver a payload to the ground while in flight. The payload delivery system will be able to carry a small package, and when lowered to the landing site, a user can prompt the payload case to open. This system will be independent of the delivery drone while the drone designed by EDT will be autonomously controlled.

1.2 Needs Statement

It can be difficult to deliver supplies to places in a timely manner. Especially in locations that are dense with trees or during dangerous weather events. It can be difficult for our current aerial systems, like helicopters and planes, or our ground delivery systems, like trucks and bikes, to reach these areas in an efficient and timely manner. For example, in 2023, as many as 30,000 residents in San Bernadino Mountains were snowed in from a massive snowstorm and rescuers were unable to send them supplies due to harsh weather conditions [1]. Rescuers found it rather challenging to send resources to these residents during the snowstorm, making the need for a more efficient payload delivery option necessary.

On a different note, it can also be difficult to deliver supplies to people who cannot leave the house, depending on the sheer demand for packages. Examine Figure 1, for example, where the number of food delivery app users has increased significantly in the US from 2015-2020. While companies like UberEats or DoorDash have been hiring people to deliver food by vehicle, there is also a need to improve food delivery output to handle the increasing number of customers.

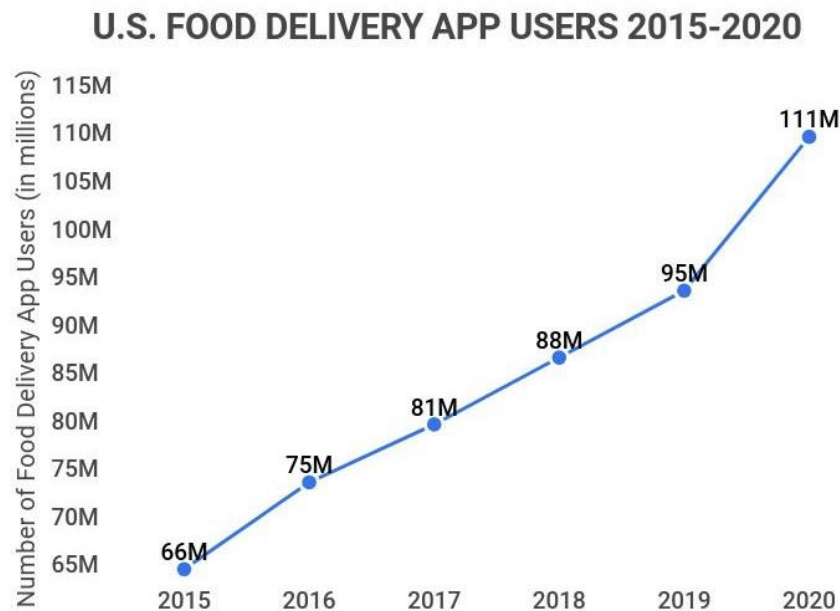


Figure 1: Chart for US Food Delivery App Users between 2015 and 2020 [2]

The Engineering Design Team is designing a system that can perform several different tasks in full autonomy for the CSU California Unmanned Aerial System Competition and possibly real-world applications. The system must be able to perform tasks like waypoint navigation, package delivery, target identification, and package recovery. It must do this through complete autonomy except for a remote pilot who can take control if the system were to malfunction. In addition, it must be a flying system, as the competition requires the system to be a rotocopter, fixed-wing vehicle, or vertical takeoff and landing vehicle.

Our senior design project focuses on the delivery system for drones. Our delivery system needs to be attached to the carrying system on the underside of the drone, and it must be able to carry a payload from one location to another. The system must be able to operate to as least 10 meters in the air. Additionally, the system will be independently controlled. It must be durable and yet lightweight to hold the payload firmly whilst the aircraft is navigating obstacles or flying along its flight path. We will be working alongside EDT to develop the delivery system while they complete their drone.

1.3 Objective Statement

The Engineering Design Team has chosen to design a quadcopter drone that utilizes computer vision for navigation and a winch for the delivery system. Since our project focuses mainly on how we can control the delivery system, we decided to design a system to handle the stability and controllability of payload delivery. In turn, it would help with delivery accuracy and mitigate any opposing forces that could impede delivery.

1.4 Background

There are many design considerations for an airdrop payload system. This section will focus on the concept behind payload drops and important considerations when designing one for drone use. Additionally, how drones combat varying forces that could impede their ability to stay on flight or conduct a delivery safely is just as important.

1.4.1 Research Survey

Two commonly used payloads systems for aerial delivery are winches and parachutes. Each provides their own benefits and concerns, but even before a package should be delivered, there should be a focus on what is being delivered. The size and weight of the payload will affect how the drone system could operate and how much it can tolerate the forces applied to it. For example, since batteries on drones are smaller, generally they won't provide enough power to keep a heavy payload in the air for a long period [3]. Large payloads will also introduce drag, slowing down the drone even more. A big part of the efficiency of a drone is making them fast.

Winch systems are devices that operate by winding a cable which creates tension to pull an object. Generally, the drum of the winch can be powered by hand, with electricity, or from hydraulics [4]. You can see winches systems used in most common areas that require heavy pulling or lifting. In vehicle recovery, it's used for pulling vehicles out of difficult terrain or towing other vehicles. In construction, it's required for lifting and moving heavy materials like loading and unloading cargo from trucks or ships. Marine application of winches is also notably important for mooring and anchoring or hauling in fishing nets. In many sectors, the technology behind the design of winches has exponentially grown where now they are better and more compact to address operational needs. A winch system with a proper winding mechanism is essential, as it will guarantee proper spooling of the drum. Designs that take this into consideration will help prolong the lifespan of the cable or rope because it reduces the friction between adjacent windings. Other design considerations include the drum, power transmission system that involve various gears and shafts, linear actuators for rope guiding, and frame housing [4]. The

dimensions of the design and materials are all dependent on operational needs or parameters. Examples include how much the winch should withstand under certain loads, or under what conditions it is operating in.



Figure 2: Drone carrying a package with a cable [5]



Figure 3: Drone with case for package and winch system in the back [6]

Components of an aerial parachute delivery can simply consist of the parachute and frame to secure the payload. However, the main issue is to consider the forces applied to the entirety of the payload. When designing a parachute system, we need to account not only for the forces from the payload but also for the forces on the parachute itself. The weight of the payload will accelerate towards the ground, providing a downward force. Then, when the parachute releases and inflates from the airflow, the canopy will generate a drag upwards. This will decrease the velocity of descent, but there are also variable conditions. The swinging motion of the payload and wind gust can change the trajectory. [7] Without a powered parachute system or steering system to guide the payload, it will land based on weather conditions.



Figure 4: Package drop with a parachute [8]

While the development of autonomous drone delivery is still ongoing, research into the assessment of the conditions of delivery zones is often overlooked. There are many different variables to make sure a delivery is done safely, such as watching out for potential hazards that could damage the package or inadvertently the drone in the process. To ensure that doesn't happen, a classification process can be used to determine if the dropping zone is a safe area or if there are other alternatives in the event it's not. Image processing is done through computer vision, which conducts the analysis of images and parameters for certain conditions to calculate if the zone is a possible option. While in flight, the algorithm will determine how much empty space there is, if motion is detected, and if there are any obstacles or hazardous objects in the background [9].

A relevant patent within this field is a coupling mechanism to attach a package hanging on a cable to a delivery drone. During delivery stage, the cable with the package is descended until the package reaches the ground, where the lack of downwards force on the mechanism allows the package to detach from the cable. In the event the mechanism fails to detach, or the cable gets stuck to an obstacle, there is a secondary severing mechanism at the drone to allow the cable to detach and the drone to fly freely. The benefits of such a system will allow for a passive detachment with no electrical components at the coupler, and any package shape can be compatible with the coupler hole attached. [10]

With current designs of drones, there are few areas that are limiting factors to successful deliveries. Accuracy decreases dramatically when systems leave controlled conditions. Environmental noise, sensor drift, and domain shift all contribute to reduced performance. When conditions change (new users, different lighting or sound environments, aging technology), false positives and false negatives become considerably more prevalent. Most inexpensive sensors have a restricted range and require regular calibration, but in practice, this maintenance is rarely performed, so performance progressively diminishes over time [9]. Battery-powered devices suffer additional constraints such as constant sensing and wireless transmission consumes a lot of power, and temperature limits cause performance to be throttled [11]. In addition, they must be lightweight for fast flight time and help with stability for maneuverability.

Building systems that work across several platforms and interface with third-party components is difficult due to varying standards and limited software development kits. This lack of compatibility restricts system design flexibility and increases development time [12]. A lot of systems come with weak encryption, default settings, and insufficient or missing secure update procedures. This creates major privacy issues and makes both data transfer and device control vulnerable to attacks [13]. It's more difficult than it should be to update existing systems. Retraining algorithms, pushing new models to devices, and keeping an eye on device health are all reasons for tedious and ineffective procedures. Improvements are slowing down, and interruption is increasing. It gets more challenging to manage device installation, security credentials, registries, and backend infrastructure [13]. A lot of user interfaces lack fundamental accessibility features and are not simple to use, which lowers usage and user confidence. Safety requirements, electromagnetic compatibility standards, and industry-specific rules are frequently disregarded or only partially addressed on the regulation side, which delays the product's rollout and creates challenges during approval.

Draganfly, DroneUp, and Arrive AI are different drone delivery companies who have recently participated in a project to get medical supplies to people in a timely manner utilizing drones [14]. In March of 2025, the department of transportation in Massachusetts

showcased the successful completion of medical delivery tests via drones, which were able to fly 10-pound packages over 1-mile trips. Draganfly offers delivery systems with either a quick-release box to release a package upon landing at a destination, or a carabiner winch for a worker at the destination to unhook and collect the package without requiring the drone to land [14]. The goal of our project will be to accomplish the same goal of delivering packages long distances with drones, but to do it with complete autonomy. The idea is to improve the already existing technology by allowing the drone to avoid obstacles and hazards on its own, which might make it more difficult for other modes of transportation to access [15].

Wing, a company that is currently leading in drone autonomy in commercial delivery, released a report detailing the potential future for autonomous drone delivery systems. They describe this technology as growing rapidly in that more industries are now adopting autonomous drones, causing the exploration of new implementations like healthcare and construction. On a similar note, Wing emphasizes the fact that drone-based payload delivery systems offer a significant reduction in carbon emissions than traditional methods of delivery. With Wing leading in the production and innovation of autonomous commercial drone deliveries, our project will prove to be another step towards autonomous payload deliveries [16].

Zipline is a company developing a drone delivery system with the original goal being a solution to deliver blood, vaccines, and other medical supplies too difficult to reach remote areas. They utilize fixed-wing aircraft like model airplanes rather than multirotor aircraft. They also use an electric launcher at distribution centers to assist in takeoff. The aircraft can fly up to 120 miles on a single charge. Payloads attached to a parachute are released at the destination. Currently, Zipline is partnering with Walmart for retail delivery, and is testing a pilot program in Arkansas [17].

Since effective payload delivery depends on the drone staying balanced and predictable in the air, the next step is to look at how the drone stabilizes itself. Before any package can be released in an accurate way, the drone needs to be able to maintain control under many different conditions (like obstacles or weather). A drones stabilization

process works by starting with the initial measurement unit (IMU), which is a component that combines a 3-axis gyroscope and a 3-axis accelerometer to measure changes in roll, pitch and yaw. The DroneZon article explains that these sensors detect very small movements caused by wind or weight shifts and send that data to the flight controller [18]. The flight controller then calculates how to correct the motion and sends the updated commands to the electronic speed controllers or ESC's. These adjust the speed of each motor so the drone can stay level, hold its current position in the air or balance out changes from the wind. These are even more important when a drone is carrying a package, the extra weight can change the center of gravity, increase drag. The IMU can adapt to these changes and give the updated info to the flight controller. Other sensors like the barometer help keep the drone's altitude steady, GPS modules that track the drone's location, or even cameras to detect visually. When all these sensors work together, they keep the drone stable even when the load varies, or the environment becomes more challenging.

While waypoints will be inputted for directions of flight path, a system utilizing computer vision to find and localize targets is required to facilitate delivery. Object detection algorithms such as You Only Look Once (YOLO) can be utilized to visually localize targets using onboard cameras and a computer for processing. A dataset containing images of example targets including edge case scenarios will be used for training a neural network for object detection. During flight, the object detection system will scan for targets on the ground, command the autopilot to fly towards the target and position directly above the target, and deliver the payload [19]. Object detection can also be utilized to detect ground obstacles such as trees and buildings, as well as searching and mapping certain objects.

To conclude, the purpose of this research survey was to highlight both the progress and the remaining challenges in autonomous drone payload delivery. While current delivery mechanisms such as winches and parachute systems offer some advantages, both are limited by their susceptibility to environmental forces like wind, which reduces delivery precision and can compromise drone flight stability and success rate of payload

delivery. While companies like Draganfly, Zipline, and Wing have demonstrated advances in commercial and medical drone delivery, these still rely on manual intervention at the destination or unguided passive release mechanisms that depend heavily on favorable conditions. The recurring theme is that performance degrades significantly outside of controlled environments. Sensor drift, battery constraints, environmental noise, and lack of cross-platform compatibility all contribute to reduced reliability in real-world deployments. These gaps directly motivate our project. By developing a system that aims to actively counteract wind disturbances and instability, we hope to undermine current delivery methods, moving one step closer to a fully autonomous and dependable drone delivery system.

1.4.2 User Survey

To get more context and understanding, we conducted a survey and interviewed our sponsor and the Engineering Design Team. The interview can be found in Appendix A and Appendix B. Our design is inspired by EDT's answers but will not be used in their drone competition. Based on our survey interviews and looking at what the competition requires, we determined that our drop system must excel in a few key areas. Speed is essential because they only have 10 minutes per mission and are graded on how many successful deliveries they make. Therefore, our system must release packages quickly to allow us to have quick turnaround times. We must be precise and delicate with deliveries because scoring is based on where the product eventually stops, not where it first hits. The delivery missions also deduct points if the payload lands too hard. This means we need smooth releases that don't cause the package to swing around, keep everything balanced during the drop, and schedule our releases correctly to account for wind and how fast the drone is flying. The system must be simple to use and dependable, particularly since they are meant to be competing in the Mojave Desert, where it is hot and windy.

Users told us they want to interact with the device as little possible to focus on drone flight operations and strong parts that can withstand dust and be used repeatedly. We also need to handle two distinct packages, a 130g beanbag and a 10cm cube, without having to rebuild anything, so the device needs to be versatile in different shapes. Everything we learnt led to the same conclusion: our mechanism must be fast, accurate, safe, and adaptable enough to function in both competition and real-world scenarios.

1.5 Objective Tree

The objectives for the drone delivery dropping mechanism were developed based on the team's analysis of the system goals and user needs. These objectives were organized into an objective tree, shown in Figure 5, to visually represent the hierarchy of design priorities.

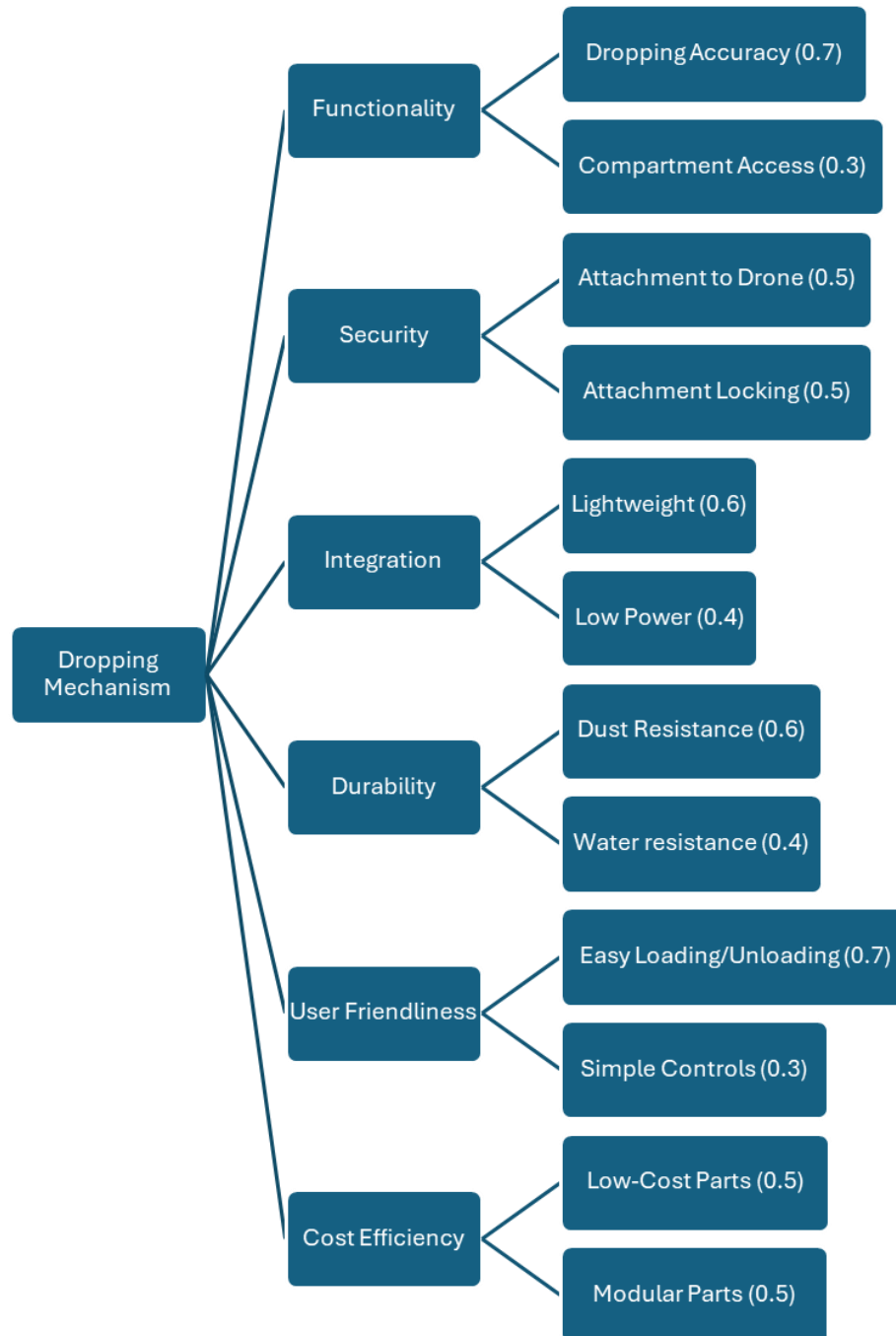


Figure 5: Objective tree

1.6 Marketing Requirements

After conducting our research and user survey and creating an objective tree, we developed marketing requirements that outline what the user wants and expects out of this product. We organized these requirements in Table 1, where we have the index of each requirement listed with a short description. They are also categorized, which makes it easier to translate into engineering requirements, which can be found in Section 2. Political, social, and cultural engineering categories do not apply for our project.

Table 1: Marketing Requirements		
Index	Marketing Requirements	MR Category
1	Should not exceed the senior design budget	Cost
2	Software system that controls the dropping accuracy should take into how environmental forces can affect delivery	Functionality
3	The hardware must be dust resistant, lightweight, and modular	Reliability
4	Should not need to take instructional input from the drone's flight system, but able to autonomously operate on its own	Interfacing Requirements with Other Systems
5	Should receive power from a LiPo battery	Power Requirements
6	Moving parts should be protected and not too easily accessed	Safety
7	Payload compartment should be easy to access	User interface
8	Should be a compact and symmetrical form factor that maintains the drone's center of gravity	Appearance
9	Should have a startup sequence	Functionality

2. Engineering Requirements

Based off the marketing requirements we made in Section 1.6, we created engineering requirements, which demonstrate the same requirements in a more abstract and verifiable criteria set. These engineering requirements, which can be found in Table 2, cover all the marketing requirements with criteria that can be measured and tested. There is a short description explaining each requirement, along with IEEE standards that relate to the requirement. See Section 6 for testing processes and results based on these engineering requirements.

Table 2: Engineering Requirements and Standards				
MR	Engineering Requirement	ER Category	Justification	IEEE Standard
1	The system will not exceed a budget of \$300	Economic	Does not specify how the budget is spent (abstract). Total price can be measured (verifiable).	ECE 396 Requirement
2	The system will guide the payload to drop the package within 2.5 meters of the center target	Performance	Does not specify how the system will guide the payload (abstract). Can measure the distance from the target center and impact (verifiable).	C-UASC Competition Requirement
3	Hardware components will be rated IP 53, be easily replaceable to which a repair can be made in less than 10 minutes, and the whole system will weigh less than 5 lbs.	Maintainability, Operational, Reliability	Does not specify how the components will be implemented with this rating or weight (abstract). Can measure the weight, repair time, and IP rating (verifiable).	IEEE 1220-2005
4	The system, when at a 30° angle from equilibrium, will stabilize in under 20 seconds autonomously	Functionality, Performance	Does not specify how the stabilization is implemented (abstract). Can test the weight, repair time, and IP rating (verifiable).	IEEE 1175.4-2008
5	The system will be powered by a 3S 11.1 V 2200 mAh LiPo	Power	Does not specific battery specifications (abstract). Can measure battery ratings (verifiable).	IEEE 1679.1-2017
6	Moving parts will be encased with a tolerance of 3mm	Operational, Safety	Does not specify how it will be designed (abstract). Can measure tolerance (verifiable).	IEEE 1936.1-2021

7	The system will allow users into the payload compartment within 2 seconds	Functionality, Performance, Usability	Does not specify how access is given (abstract). Can measure time (verifiable).	IEEE 29148-2018
8	The system will not exceed 40 cm in length, 30 cm in width, and 20 cm in height so that it doesn't affect drone flight	Operational, Manufacturability	Does not specify how it affects drone flight (abstract). Can measure dimensions (verifiable).	IEEE 15288-2024
9	The system will have an initialization sequence that will last no more than 15 seconds	Functionality, Operational	Does not specify specific sequence (abstract). Can measure how long (verifiable).	IEEE 2050-2018

3. Design Alternatives

Before tackling our problem and addressing the design alternative process, we created a concept map to organize our options. This map can be seen in Appendix C. Design alternative 1, 2, and 3 are possible alternatives to tackle the problem of controllability and stability of payload delivery. Each design alternative tackles the problem of stability and accuracy through different methods.

3.1 Design Alternative 1

Design Alternative 1 focuses on designing a case or box in which the payload will be loaded. This case will have fans pushing forward, left, and right. The system will have sensors to know the angle and rotation of the payload box and adjust according to keep it level. There will also be sensors below to judge the distance to the ground.

Table 3: Level 0 for Design Alternative 1	
Module	Payload Case Guidance
Inputs	Angular Rotation Sensor Distance Sensor Power, 22.2 V DC
Outputs	Fan Adjustment Feedback Door hatch signal for mechanical package release
Functionality	It receives data from the angular rotation sensor and transmits it to the microcontroller. Fans will operate whenever and at whatever speed needed to combat swinging of the payload. At the same time, data from the distance sensor will allow the system to know if a user wants to open the payload compartment. The box has a door hatch mechanism which will open and release the payload.

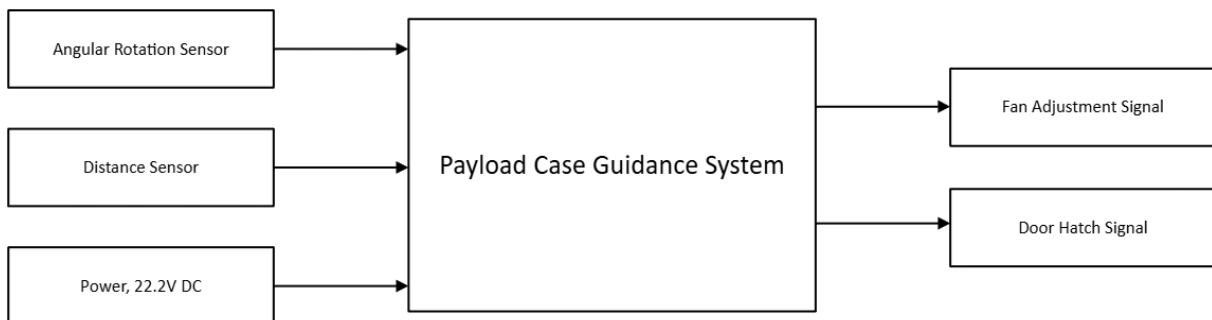


Figure 6: Level 0 schematic for Payload Case Guidance System

3.2 Design Alternative 2

Design Alternative 2 focuses on the controllability of the drone while payload is being sent to the ground. The payload will be loaded into a case. The drone and payload case will have its own Inertial Measurement Unit or IMU. The payload acts as a pendulum, and the sensor modules will allow the drone to calculate trajectory and adjust the propellers on the drone accordingly.

Table 4: Level 0 for Design Alternative 2	
Module	Pendulum Guidance
Inputs	Inertial Measurement Unit Sensor from Drone Inertial Measurement Unit Sensor from Payload Case Power, 22.2 V DC
Outputs	Signals to drone propellers
Functionality	The IMU on the drone will send data such as altitude, linear acceleration, and velocity to the microcontroller. At the same time, the IMU on the payload case will send data such as payload swing angle and payload swing rate. The controller will calculate and estimate adjustments needed to reduce swings. The flight controller will send signals to the propellers accordingly.

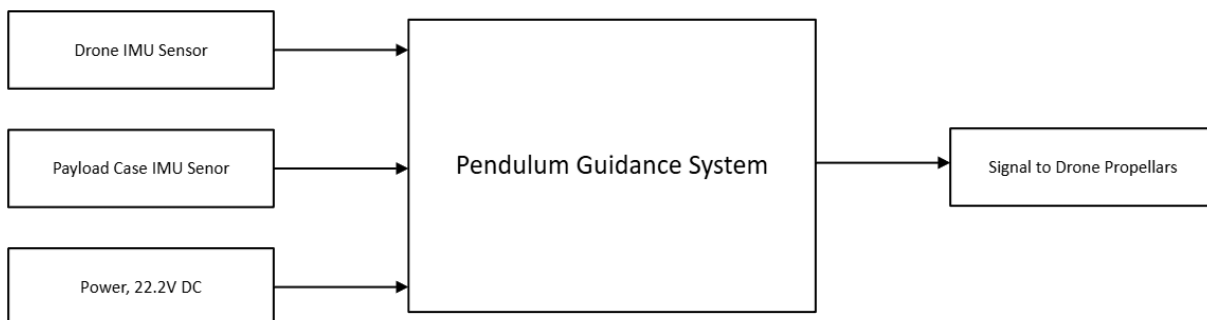


Figure 7: Level 0 schematic for Pendulum Guidance System

3.3 Design Alternative 3

Design Alternative 3 focuses on active winch control. The drone will already have a winch mechanism in place that will lower or pull up when signaled to. Changing the cable length at the right time can reduce the swing energy potential or prevent it from growing. This will actively cancel the swing motion. The payload will be loaded into a case which will have its own Inertial Measurement Unit or IMU. The sensor module will allow the drone to calculate trajectory and adjust the winch controls.

Table 5: Level 0 for Design Alternative 3	
Module	Winch Guidance
Inputs	Inertial Measurement Unit Sensor Power, 22.2V DC
Outputs	Signals to winch to shorten or lengthen cable
Functionality	The IMU on the payload case will send data such as payload swing angle and payload swing rate to the microcontroller. The controller will calculate and estimate adjustments needed to reduce swings. The flight controller will send signals to the winch accordingly.

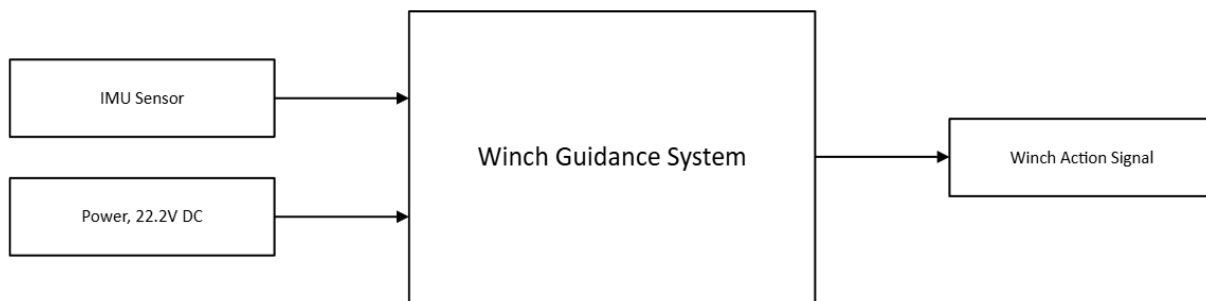


Figure 8: Level 0 schematic for Winch Guidance System

3.4 Differences and Similarities

Between the three main design alternatives, there are some differences and similarities we can see. Design Alternative 1 works around with the payload itself and how we can stabilize it during delivery. Design Alternative 2 works with adjusting the drone based on how the payload is behaving. Design Alternative 3 works similarly to Design Alternative 2, on how adjustments are made from how the payload is behaving, but it's done through the winch mechanism. In terms of components, Design Alternative 1 uses an angular rotation and distance sensor. However, Design Alternative 2 and 3 mainly uses an Inertial Measurement Unit. This is key in helping the drone calculate the adjustments needed to counteract any forces that could affect the delivery. In terms of speed, Design Alternative 3 will be the slowest because lengthening and shortening the cable throughout the descent of the payload will take a larger amount of time. Shortening the cable will mean that it will backtrack during the descent. Design Alternative 1 and 2 just involve adjustments to fans and propellers which can happen a lot quicker. In terms of safety, Design Alternative 1 might have the most concerns. The fans on the payload case can malfunction and cause more swaying or force that would affect the drone's ability to stay hovering. This design is more actively involved with controlling the payload, so it could be more likely to crash compared to the other design alternatives.

3.5 Design Alternative Changes

Design alternatives have been updated since preliminary designs were complete. The main source of power will come from an 11.1V battery. The team determined that an 11.1V LiPo battery will be sufficient to power the entire system. This will help cut back on cost and weight. Regarding Design Alternative 1, an altitude sensor and a push button have been added to the design as an input into the system. The main sensor used in this sensor will be an inertial measurement unit that contains an accelerometer and gyroscope.

4. Decision Matrix

This section will focus on selecting the best design alternative for our project. The following are our list of criteria: development cost, contribution to ECE education, design complexity, technical knowledge, and time allocation. This list of design criteria was derived from several factors that our team needs to consider when designing the payload case. We have a certain budget and can't afford to build an expensive system, so development costs will be a restricting factor. We chose contribution to ECE education, design complexity, and technical knowledge because these will play the biggest role as it addresses our ability to handle the project and how valuable it can be for our careers. For time allocation, we have a deadline to finish the project, and it needs to be presented at EXPO. Therefore, it needs to be completed in a feasible amount of time. Each team member conducted their own pair-wise comparison on a list of agreed-upon criteria. The scale is as follows: 1 = equal, 3 = moderate, and 5 = strong, 7 = very strong, 9 = extreme. The scale represents a row being x times more important than the column. Team members' individual pair-wise comparisons are in Appendix D-I. Table 6 shows the matrix which has the average of the members' ratings with the scale considered.

Table 6: Final Pair-Wise Comparison Matrix							
	Development Cost	Contribution to ECE Education	Design Complexity	Technical Knowledge	Time Allocation	Geometric Mean	Weights
Development Cost	1	5	3	1	3	2.14	0.37
Contribution to ECE Education	1/5	1	1	3	3	1.12	0.20
Design Complexity	1/3	1	1	1	5	1.11	0.19
Technical Knowledge	1	1/3	1	1	3	1.00	0.17
Time Allocation	1/3	1/3	1/5	1/3	1	0.37	0.07
Total							1.00

Next, as a team, we rated each design alternative relative to the criteria and the average between the team members is compiled. These ratings are found below in Table 7. The scale is as follows: 1 = does not meet the criterion, 3 = partially meets the criterion, and 5 = completely meets the criterion.

Table 7: Alternatives Relative to Criteria					
	Development Cost	Contribution to ECE Education	Design Complexity	Technical Knowledge	Time Allocation
Design Alternative 1	3	4.3	4.2	4.3	3.5
Design Alternative 2	5	4.3	3	1.7	2
Design Alternative 3	4.3	2.3	2	2.2	3.5

Now with the data we've collected, we can use it to create a decision matrix to determine which design alternative is the best choice. The scoring can be found in Table 8.

Table 8: Decision Matrix				
	Weight	Design Alternative 1	Design Alternative 2	Design Alternative 3
Development Cost	0.37	3	5	4.3
Contribution to ECE Education	0.20	4.3	4.3	2.3
Design Complexity	0.19	4.2	3	2
Technical Knowledge	0.17	4.3	1.7	2.2
Time Allocation	0.07	3.5	2	3.5
Score		3.74	3.71	3.05

With the information we gathered, we've decided that Design Alternative 1, Payload Case Guidance System, is our best design alternative. Design Alternative 1 had the highest score among the three design alternatives. On average, the criteria in Design Alternative 1 have either the same or the highest score, which shows how it's not lacking in any category, besides the development cost.

5. Preliminary Design

Here we present some preliminary designs for our design alternative. The core of the system will involve one inertial measurement, the microcontroller, and motors. Additionally, the control system is heavily reliant on software algorithms to compute and send signals to the motors to adjust. The software subsystems will involve calls to our inertial measurement units to send data from their sensors to interact with the microcontroller to make calculations and send adjustment instructions.

5.1 Level 1 Schematic

Look at the Level 1 schematic; our battery is supplying 11.1V, powering our motors, and it is sent through a buck converter to provide appropriate voltages to the other different subsystems. The accelerometer and gyroscope sensors on the IMU will send data to the flight controller and with control algorithms send signals to electronic speed controllers for the propellers to make some adjustments. Figure 9 expands and shows the components involved in the system.

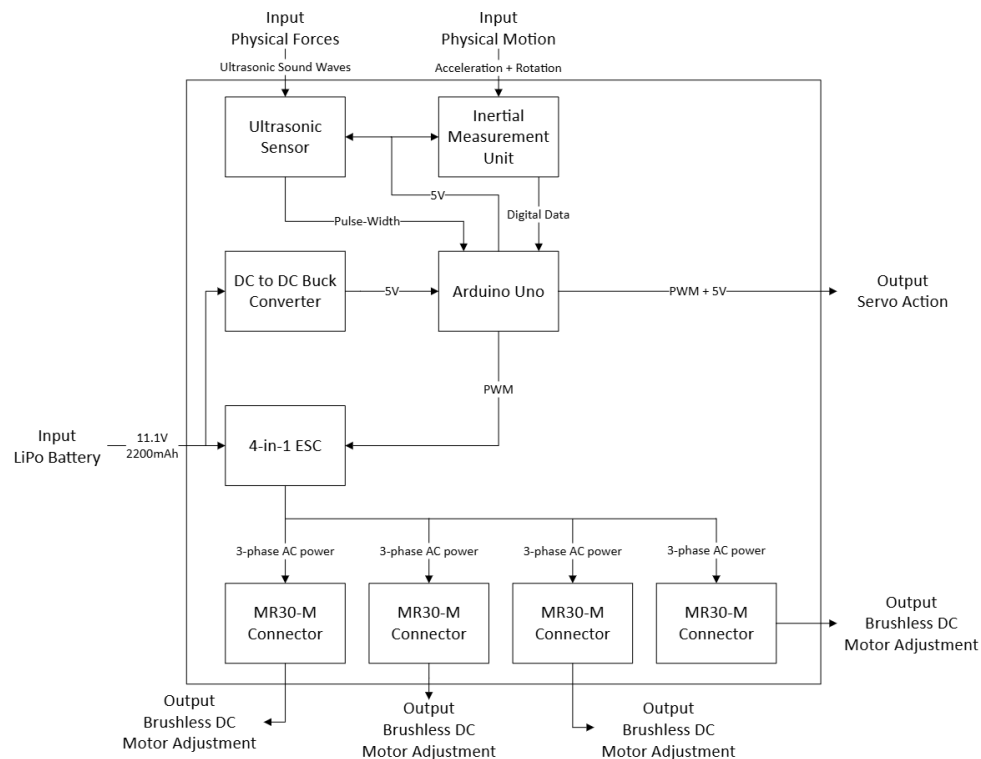


Figure 9: Level 1 schematic for Pendulum Guidance System

5.2 Circuit Design

Figure 10 goes into the preliminary circuit design for the Pendulum Guidance System. The two main hubs of the system, the 4-in-1 electronic speed controller and the Arduino, are the central components. The Arduino will house all the sensors required to compute and signal adjustments, and the 4-in-1 electronic speed controller is the hub for the motors.

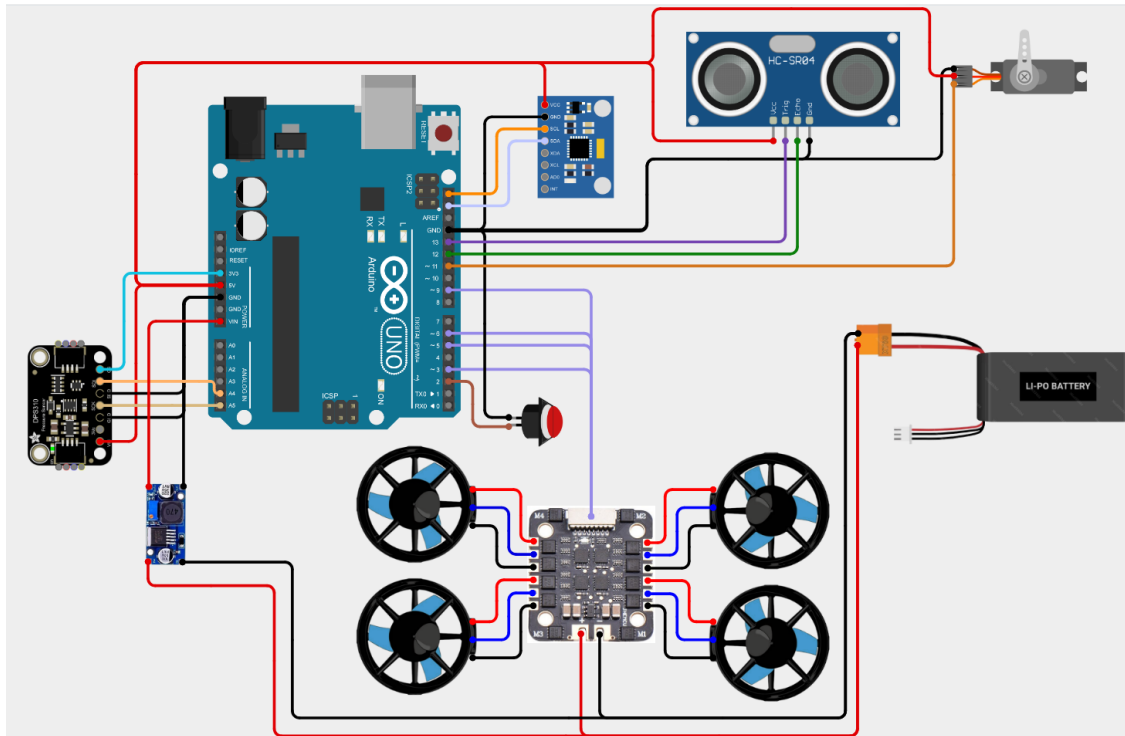


Figure 10: Preliminary Circuit Design

5.3 Software Design

Designing a control system for this design will involve writing software in Arduino's Integrated Development Environment (IDE) that will allow the microcontroller to take in information from various sensors and compute different adjustments. These signals will be sent to help the payload case stay stable and level during transit and delivery. Figure 11 is a flowchart for the subroutines control system that is meant to help stabilize the payload case during transit and delivery. The control system has two algorithms that work side by side. The control/stabilization algorithm will call the IMU to collect acceleration and rotations data provided by the accelerator and gyroscope. The algorithm will then determine if the payload case is off axis, not in a balance position, and if it is, it will send

signals to motors to make some adjustments. This is how it will correct the off-balance the payload case might arise during delivery. In addition, acceleration data will help prevent oscillations. The altitude will also be called to send data so that the payload case can keep track of how high it is in the air. Doing so will also allow the algorithm to restrict when the motors should be operating for safety concerns. The latch compartment algorithm will call the ultrasonic sensor to collect data, so the case can determine if a hand wave gesture was made. That is the only way that would allow the end user to open the payload compartment to retrieve or load a package.

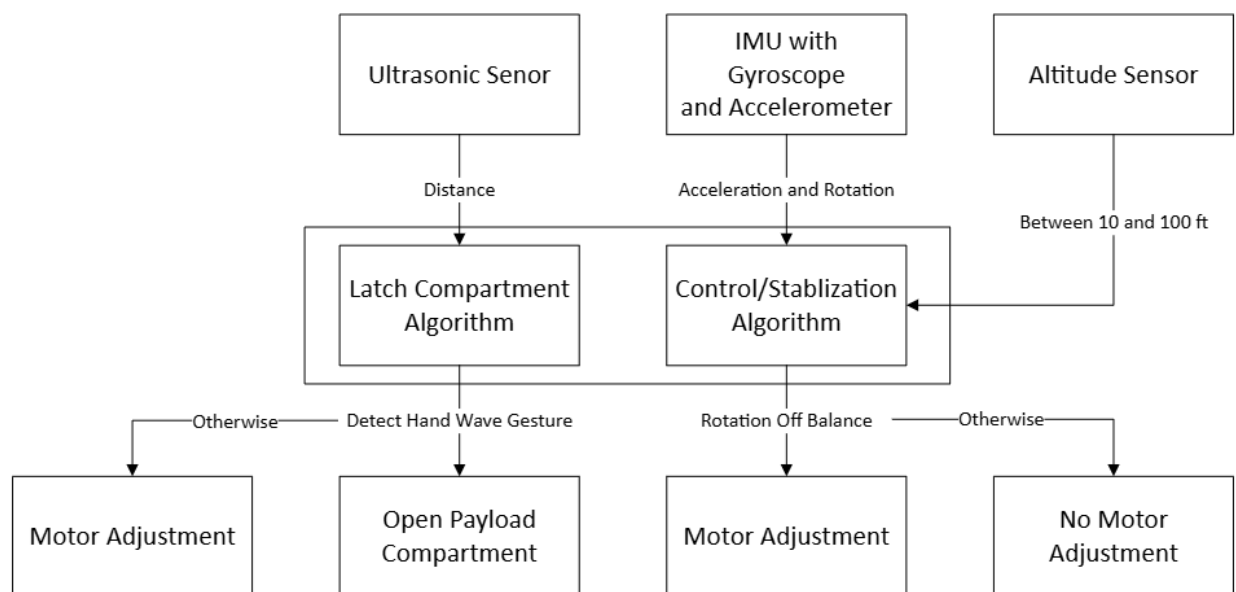


Figure 11: Flowchart for Control System

6. Testing

This next section is our documentation that describes what testing (unit, integration, and acceptance) we conducted and shows the results of those tests. There were three different unit tests. The first was testing that the IMU was able to measure pitch and roll angle and update as changes were made. The second was testing that our ultrasonic sensor was able to measure distance and update as changes were made. The third was testing that our 4-in-1 electronic speed controller allowed our motor motors to spin clockwise and counterclockwise. There were three different integration tests. The first integration test was having a push button to be a power button for the system, indicating the start of the initialization sequence. The second test was integrating the ultrasonic sensor with the servo motor. The third test was integrating the 4-in-1 electronic speed controller and motors with the IMU. The acceptance testing involved testing how long it took for the case to reach an equilibrium when pushed away and the accuracy of payload drops.

6.1 Unit Testing

The step up for testing the IMU can be seen in Appendix K. We were able to acquire the initial pitch and roll values as those will be required to compute the pitch and roll correction based on error from initial values. It will also constantly update as long as the system is active, so if there are changes to pitch and roll, the system will be able to keep track. We know that it was operating correctly because as you tilt the IMU in a certain direction, the serial monitor will print the updates accordingly.

The step up for testing the ultrasonic sensor can be seen in Appendix L. For this unit testing we wanted to measure the distance from the sensor in centimeters which can be seen in the serial monitor. We were able to get the sensor to constantly update the distance alongside the IMU. It was able to detect when there was a hand wave motion denoted by the 5cm in the serial monitor showing. This means that it was reading the distance from the folder and as a hand was motioned across the sensor it read that change of distance. This unit testing worked as required for our system.

The step up for testing the 4-in-1 electronic speed controller and the motors can be seen in Appendix M. Testing was done alongside integration testing that will be explained shortly after this. To test that the motors were able to spin clockwise and counterclockwise, we had tape in front of the motors. This would show the change of thrust direction that the motors provided when switching rotation. The tape would be sucked into the motor with the motor spinning one direction and the tape pushing away from the motor when the motor rotated in the opposite direction. This unit testing worked as required for our system.

6.2 Integration testing

The setup for the push integration can be seen in Appendix K. Once the button is pressed, an initialization sequence is started to calibrate the IMU and get the initial pitch and roll angle values. This can be seen on the serial monitor. We were able to successfully integrate this together. The setup for the servo integration can be seen in Appendix L. When the ultrasonic sensor recognizes that a hand wave motion has been detected by observing the change in distance, the servo motor activates. The first activation should make the servo rotate clockwise by 90 degrees, and the second activation should make the motor spin the servo counterclockwise by the same 90 degrees. This mechanism will act as the opening and closing of the latch door to access the payload compartment. We were able to successfully integrate this together. The setup to integrate the 4-in-1 electronic speed controller and motors with the IMU can be seen in Appendix M. The rotation of the motors should be dependent on the orientation of the IMU. Based on the initial pitch and roll values obtained in the initialization sequence, the motors will rotate in a specific direction based on which direction the pitch or roll has deviated from. There are four motors, two will correct pitch, and the other two will correct roll. So, for example if there is an overcorrection, the two motors will rotate clockwise to bring it back down and if there is an under correction the same motors will rotate counterclockwise to bring it back up. We were able to successfully integrate this together.

6.3 Acceptance Testing

We tested a control group against our control system. There were two tests. The first test showed how long the payload case took to reach a balanced/stable position. The second test showed drop accuracy. Accuracy was measured based on how far the cases landed from the center of our target zone. In addition, we tested the case in different orientations to simulate different paths the case could take while in the air during a live delivery.

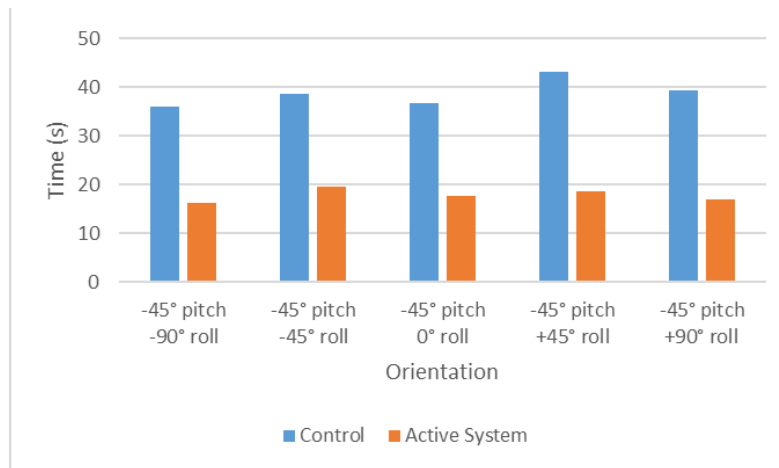


Figure 12: Time to Stabilization

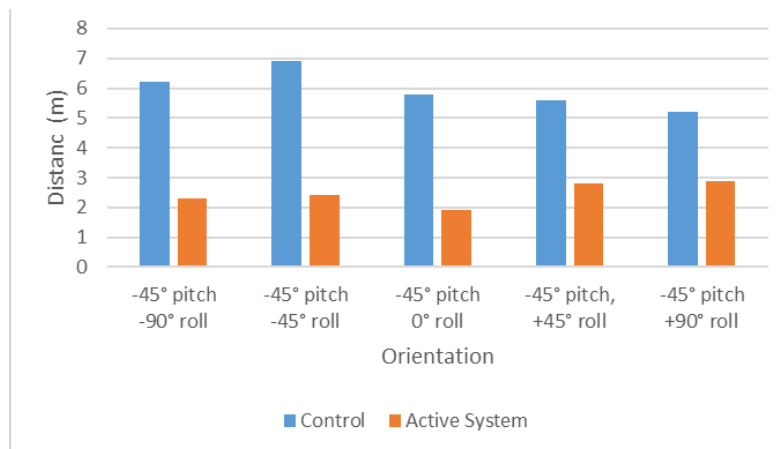


Figure 13: Drop Accuracy (20 ft in 15 sec)

We noticed that with our control system, the payload case is on average more than 53.9% faster and 58.7% more accurate than a payload case without any active stabilization methods. This proves that our control system works and has a positive impact. In addition to raw testing, the testing below shows the acceptance testing to verify if we were able to meet Engineering Requirement.

Table 9: Acceptance Testing for Time Stabilization

Table 9: Acceptance Testing for Time Stabilization						
Test Writer: Jeremy Vo						
Test Case Name:		Time Stabilization Testing			Test ID #:	TS2026
Description:		The system, when at a 30° angle from equilibrium, will stabilize in under 20 seconds autonomously.			Type:	☐ white box ☒ black box
Set up:						
Step	[Insert Input/Parameter]	Expected Result		Pass	Fail	Comments
1	Set case at a 30° angle away from pitch equilibrium	Case will stabilize in under 20 seconds		☒	☐	Averaged 17.83 seconds
2	Set case at a 30° angle away from roll equilibrium	Case will stabilize in under 20 seconds		☒	☐	
Overall test result:				Pass		

Table 10: Acceptance Testing for Payload Compartment Access

Table 10: Acceptance Testing for Payload Compartment Access							
Test Writer: Jeremy Vo							
Test Case Name:		Payload Compartment Access			Test ID #:		US2026
Description:		The system will allow user to the payload compartment within 2 seconds			Type:		☐ white box ☒ black box
Set up:							
Step	[Insert Input/Parameter]	Expected Result		Pass	Fail	Comments	
1	Wave hand across ultrasonic sensor	Servo will trigger and open door compartment within 2 seconds and remain open		☒	☐	N/A	
2	Wave hand across ultrasonic sensor again	Servo will trigger and close compartment and remain closed		☒	☐	N/A	
Overall test result:				Pass			

Table 11: Acceptance Testing for Initialization Sequence Timing

Test Writer: Jeremy Vo						
Test Case Name:		Initialization Testing			Test ID #:	IS2026
Description:		The system will have an initialization sequence will last no more than 15 seconds			Type:	☐ white box ☒ black box
Set up:						
Step	[Insert Input/Parameter]	Expected Result	Pass	Fail	Comments	
1	Push the push button	Each propeller turns on consecutively. After each propeller activates briefly, the system will begin its calibration state, this last 10 seconds	☒	☐	N/A	
Overall test result:			Pass			

Table 12: Acceptance Testing for Time Stabilization

Test Writer: Jeremy Vo						
Test Case Name:		Accuracy Testing			Test ID #:	AT2026
Description:		The control system will guide the payload to drop the package within 2.5 meters of the center target			Type:	☐ white box ☒ black box
Set up:						
Step	[Insert Input/Parameter]	Expected Result	Pass	Fail	Comments	
1	Set case at a 30° angle away from pitch equilibrium and descend the payload case	Should land within 2.5 meters of the center target	☒	☐	Done on a stairwell 20 feet above ground with a rope attached to the claw mount to drop down. Averaged 2.46 meters.	
2	Set case at a 30° angle away from roll equilibrium and descend the payload case	Should land within 2.5 meters of the center target	☒	☐		
Overall test result:			Pass			

Table 13: Verifying All Engineering Requirements

Engineering Requirement	Acceptance
The system will not exceed a budget of \$300.	Meet as the bill of all materials was \$279.77.
The system will guide the payload to drop the package within 2.5 meters of the center target.	Meet as the average distance from center target was 2.46 meters.
Hardware components will be rated IP 53, be easily replaceable to which a repair can be made in less than 10 minutes, and the whole system will weigh less than 5 lbs.	Meet as components are protected and mostly sealed inside the case. Different subsystems are connected via XT connections allowing replacements if needed. The payload case also weighed 4.98 pounds.
The system, at a 30° angle from equilibrium, will stabilize in under 20 seconds autonomously.	Meet as the average time took 17.83 seconds but at an even greater angle of 45° from equilibrium.
The system will be powered by a 3S 11.1 V 2200 mAh LiPo	Meet as we were required to use a 3S LiPo battery which is at a nominal 11.1 V.
Moving parts will be encased with a tolerance of 5mm.	Meet as fan grills were designed with a tolerance of 3mm ± 0.2mm given 3D printer.
The system will allow users into the payload compartment within 2 seconds	Meet as payload compartments can be accessed instantly as you trigger the ultrasonic sensor.
The system will not exceed 40 cm in length, 30 cm in width, and 20 cm in height so that it doesn't affect drone flight	Meet as the case has dimensions of 36 cm in length, 24 cm in width, and 14.75 cm in height.
The system will have an initialization sequence will last no more than 15 seconds	Meet as initialization sequence last 10 seconds.

7. Validation

We could not validate our product to EDT as their competition is in June, and by the time this project needed to be turned in for ABET, the drone winch system was not yet completed.

8. Risk Analysis

The main safety risk in our design was the propeller and motor mounts. See Appendix M for the original design of the motor mounts and Appendix N for the updated model. The motors can have high rpms and can cause serious harm to a user. Not only that, but serious damage can be done to the motors if not protected from other elements. Fan grills were designed to prevent the potential risk of users getting caught by the propeller and protect the motors from large objects. The 4-cable attachment to the claw mount was also designed to be long enough away from the top of the payload case so that the door compartment wouldn't catch on the door. If it did, it would cause the servo to not open all the way and close back up. A failure in our design came through when we integrated our door piece to the servo. When the door is down at the rest, the servo isn't strong enough to hold it up. As a result, the door would collapse inward into the payload compartment. Little standoff lips were designed so that the door could rest on top of it without falling inwards. Another huge failure was the flexibility of the material we chose to build the case out of. Payloads with serious weight could crack even snap the frame during transit. To fix this, we design brackets to hold the frame together in weak spots around the case. This made the case more firm and rigid. A detailed obstacle and error documentation can be found in Appendix P.

9. Final Design

Our final prototype compiles all the research, testing, and analysis into one functional system. A photo of our final design with the drone developed by EDT can be found in Appendix N. The final design consists of a push button to signal to power on/off the system and start the initialization or shut down sequence. The Arduino houses the altitude sensor and inertial measurement unit which during the initialization sequence stores initial altitude, roll, and pitch data. There are four motors that are integrated into a hub together through a 4-in-1 electronic speed controller. Through the 4-in-1 ESC and signaled by the Arduino, the motors will operate when needed to correct the orientation of the payload based on the error calculated from the initial data. Two motors operate alongside each other to correct one angle. A pair of motors will correct roll, and the other pair will correct pitch. Then, there is an ultrasonic sensor that is used to detect a hand wave motion and when triggered will signal the servo to open/close the latch for door of the payload compartment. Everything is powered by 3S 11.1V LiPo Battery at 2200mAh. A draft user manual, detailing some user instructions and product information can be found in Appendix J. A task allocation and timeline detailing the organization of the project timeline can be found in Appendix O.

9.1 Bill of Materials

The cost breakdown of this project can be seen below as a list of all the hardware and its associated costs. As we source our hardware from well-known manufacturers and suppliers, our pricing will be relatively comparably and competitive. Table 14 lists each part, their cost, the seller, quantity, and the total cost with respect to the quantity.

Table 14: Parts List and Cost				
Part	Cost	Store	Quantity	Total
BNO055 9DOF IMU	\$34.95	Adafruit	1	\$34.95
Continuous Rotation Servo	\$11.59	Adafruit	1	\$11.59
UBEC DC/DC Step-Down Converter	\$9.95	Adafruit	1	\$9.95
DPS310 Altitude Sensor	\$6.95	Adafruit	1	\$6.95
Arduino Uno Rev3	\$20.7	Arduino	1	\$20.7
Proto Shield Rev3 Kit	\$14.10	Arduino	1	\$14.10
3-Blade 3D Propeller	\$3.99	GetFPV	2	\$7.98
iFlight XING-E Pro 2207 Motor (1800KV)	\$22.49	GetFPV	4	\$89.96
XILO Stax V3 F4 50A AM32 3-6S 4-in-1 ESC	\$64.99	GetFPV	1	\$64.99
Ultrasonic Distance Sensor (HC-SR04)	\$5.25	Sparkfun	1	\$5.25
2200mAh 3s (11.1v) 35C LiPo Battery	\$13.35	Altitude Hobbies	1	\$13.35
Total Cost				\$279.77

10. Contribution of Work

Individual contribution of work from each member.

Jeremy: Worked heavily with the software/control algorithm. I built the control algorithm script, and this included doing the fine tuning of the control system and testing phase. The testing included unit testing, integration testing, and acceptance testing for some of the engineering requirements. I organized the layout of our circuit design and how each subsystem should be wired and integrated together. Additionally, I laid out the initial design of the prototype, where the different subsystems should go and how many fans we needed. I also helped with building the physical payload case; screwing the frame and mounts together.

Stuart: Helped build the physical payload case by soldering holes into acrylic panels for fan mounts and screwing system components together. Researched the best material to make the payload case; as well as helped determine design alternatives. Researched how the payload case would be attached to the drone, and forces based on those orientations.

Samuel: I worked heavily on the payload case design and hardware. I crafted the first cardboard prototype to integrate all the subsystems for testing. I was responsible for designing the 3D model for the four motor and propeller mounts and the drone claw handle. Additionally, I used my 3D printer to print all the PLA parts that both my team members and I created. Later, I would go on to help build and assemble the final prototype alongside the rest of the group members. I was also responsible for soldering hardware. This includes soldering XT connectors for each motor, soldering the altitude and gyroscope sensor onto the protoshield, and soldering together a power system to connect the battery to the microcontroller and ESC.

Kyle: I worked extensively on the arm mount that attaches to the servo for our door latch. I measured and made multiple 3D model designs in SolidWorks and prototypes to make sure the angle that the door opens is completely flush with the rest of the box and that it

also does not take up any unnecessary space in our payload box. I also helped with the assembly of our final acrylic box with attaching and screwing the fans to the acrylic panels as well as with soldering holes through the acrylic so we could screw in the door lid to the 3D printed arm.

John: I contributed heavily to the physical assembly and hardware design of the payload case. Additionally, I assisted with the physical fabrication process, specifically working with the acrylic plates to properly construct, mount, and assemble the final external housing for the drop mechanism.

Adrian: Did research and selected components for power and propulsion, and a wiring plan for secure electrical connections while ensuring ease of serviceability. Designed and CAD modeled the acrylic plates for the payload case and the 3D printed brackets holding the pieces together. Also, I soldered the ESC and set up its firmware. Lastly, assisted with assembling the prototype.

11. Conclusion

During crucial drone deliveries, where time, efficiency, and accuracy matter, the AirDrop Transport Companion strives to make autonomous drone package delivery a more viable option. Our senior design team, along with the Engineering Design Team, worked to create a product that is stable and accurate by using a propeller-based PD control system that counteracts swaying and controls the descent of the payload case. Our product fits the needs of the marketing and engineering requirements for such a system, including size, weight, accuracy, tolerance, and speed. With multiple design alternatives and iterations, we settled with a choice that both fits the needs of the user and the engineering capabilities of our group members. In terms of testing, our final prototype was able to stabilize in about 17.83 seconds when released at 30° away from equilibrium. When waving your hand in front of the ultrasonic door, it would open and close as expected. When testing for landing accuracy at a height of 20 feet, we were able to land within 2.46 meters of the target on average. Our system performed about 53.9% faster and 58.7% more accurate than a system without a control algorithm. Finally, the payload case weighs 4.98lbs, uses an 11.1V 2200mAh power supply, and fits the budget at \$279.77. With these results, we are confident that our senior design project is a big step towards improving autonomous payload delivery and our own individual engineering experience. There are many things we can improve upon in the future, including research, additional planning, and organizing. There were a couple of problems regarding ordering parts and integrating parts with the whole system. Individual takeaways from this project can be found in Appendix Q. Regardless, it was a learning experience for the whole team, and every member can confidently move forward knowing they have more engineering experience than they had before.

References

[1]

Venturi, “At Least 13 Deaths In Mountains During Blizzard Of ’23,” SBCSentinel, <https://sbcsentinel.com/2023/09/at-least-13-died-in-mountains-during-winter-blizzard/> [Accessed Oct. 19, 2025]

[2]

J. Flynn, “18+ Food Delivery Statistics [2023]: Online ordering industry numbers you need to know,” Zippia, https://www.zippia.com/advice/food-delivery-industry-statistics/#Online_Food_Delivery:_Consumer_Ordering_Preferences [Accessed Oct. 19, 2025]

[3]

E. Frachtenberg, “Practical Drone Delivery,” vol. 52, no. 12, pp. 53–57, 2019, doi: 10.1109/MC.2019.2942290.

[4]

H. A. Yasin, M. H. A. Awan, F. Khattak, and A. Hannan, “Design of Winch System with Automated Cable Winding Mechanism,” presented at the 2024 21st International Bhurban Conference on Applied Sciences and Technology (IBCAST), 2024, pp. 12–17. doi: 10.1109/IBCAST61650.2024.10877206.

[5]

Adobe Stock. Accessed Oct. 19, 2025. [Photo] Available: https://stock.adobe.com/images/drone-carrying-supplies/290559386?prev_url=detail

[6]

Adobe Stock. Accessed Oct. 19, 2025. [Photo] Available:

https://stock.adobe.com/images/drone-transporting-a-first-aid-to-the-city/633544986?prev_url=detail

[7]

A. Shankar, S. Elbaum, and C. Detweiler, “Towards Aerial Recovery of Parachute-Deployed Payloads,” presented at the 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2018, pp. 4700–4707. doi: 10.1109/IROS.2018.8594082.

[8]

Adobe Stock. Accessed Oct. 19, 2025. [Photo] Available:

https://stock.adobe.com/images/package-drop-via-drone-a-drone-dropping-a-package-attached-to-a-parachute-ensuring-safe-delivery-to-a-remote-location/939309107?prev_url=detail

[9]

A. Alsawy, A. Hicks, D. Moss, and S. McKeever, “An Image Processing Based Classifier to Support Safe Dropping for Delivery-by-Drone,” presented at the 2022 IEEE 5th International Conference on Image Processing Applications and Systems (IPAS), 2022, pp. 1–5. doi: 10.1109/IPAS55744.2022.10052868.

[10]

J. Hayden, J. Rinaldi, M. Sweeny, “Package Delivery Mechanism,” U.S. Patent 11840333B2, Dec. 12, 2023. [Online] Available: <https://patents.google.com/patent/US11840333B2>.

[11]

W. Shi, J. Cao, Q. Zhang, Y. Li, and L. Xu, “Edge Computing: Vision and Challenges,” *IEEE Internet of Things Journal*, vol. 3, no. 5, pp. 637–646, Oct. 2016, doi:

10.1109/JIOT.2016.2579198. https://cse.buffalo.edu/faculty/tkosar/cse710_spring20/shi-iot16.pdf. [Accessed: Sep. 28, 2025]

[12]

E. Lee, Y.-D. Seo, S.-R. Oh, and Y.-G. Kim, "A Survey on Standards for Interoperability and Security in the Internet of Things," *IEEE Communications Surveys & Tutorials*, vol. 23, no. 2, pp. 1020–1046, 2021. <https://ieeexplore.ieee.org/document/9381989> [Accessed: Sep. 29, 2025]

[13]

M. Fagan, K. Megas, K. Scarfone, and M. Smith, "IoT Device Cybersecurity Capability Core Baseline," NISTIR 8259A, National Institute of Standards and Technology, May 2020. [Online]. <https://doi.org/10.6028/NIST.IR.8259A> [Accessed: Sep. 29, 2025]

[14]

Draganfly Innovations Inc., "Precision Delivery System," Draganfly, 2025. [Online]. Available: <https://draganfly.com/precision-delivery-system/>. [Accessed: Oct 10, 2025].

[15]

Commercial UAV News Staff, "MassDOT Successfully Pilots Medical Drone Delivery Project," *Commercial UAV News*, Mar. 17, 2025. [Online]. Available: <https://www.commercialuavnews.com/massdot-successfully-pilots-medical-drone-delivery-project>. [Accessed: Sep. 16, 2025]

[16]

M. McNabb, "Wing Releases 'Beyond the Aisle' Report: Key Findings on the Future of Drone Deliveries," *DroneLife*, Sep. 19, 2024. [Online]. Available: <https://dronelife.com/2024/09/19/wing-releases-beyond-the-aisle-report-key-findings-on-the-future-of-drone-deliveries/>. [Accessed: Sep. 16, 2025]

[17]

T. Stevens, “Don’t call it a drone: Zipline’s uncrewed aircraft wants to reinvent retail,” *Ars Technica*, <https://arstechnica.com/gadgets/2025/04/dont-call-it-a-drone-ziplines-uncrewed-aircraft-wants-to-reinvent-retail/> [accessed Oct. 5, 2025].

[18]

Corrigan, F., “Drone Gyro Stabilization, IMU And Flight Controllers Explained,” *DroneZon*, May 7 2020. <https://www.dronezon.com/learn-about-drones-quadcopters/three-and-six-axis-gyro-stabilized-drones/>. [Accessed: Nov. 15 2025].

[19]

V. Godase, S. Modi, V. Misal, and S. Kulkarni, “Real-Time Object Detection for Autonomous Drone Navigation Using YOLOv8,” *MAT Journals*, Jul. 15 2025. https://www.academia.edu/download/124126736/_17_27_Real_Time_Object_Detection_for_Autonomous_Drone_1_.edited.pdf. [Accessed: Nov. 15 2025]

Appendix

Appendix A

Interview Questions for Sponsor

1. How do you expect us to work in collaboration with the EDT team?

“Refer to the EDT team for collaboration. Whether you should work on part of the drone or the entirety of the drone, all depends on a plan that will be communicated between the senior design group and the EDT team. The senior design group will not be expected to complete the entire drone since competition is in June, which is well outside the range of the senior design timeline.”

2. Since we’ll be working with you and the EDT team, what types of questions or issues would you prefer us to bring to you, and which should we go to the EDT for?

“The sponsor contributed very little to the technical design team. They helped with administrative tasks and paperwork. Technical questions are welcome (control, programming, etc.). Most of the work is expected to be completed by EDT and Senior Design with infrequent sponsor intervention.”

3. How and what qualifications would this project need to meet for this project to be considered a success?

“The drone should fly for a certain amount of time and energy based off guidelines. Design the drone to fit those guidelines and goals. Consider the fact that June is when the competition is, and the project must be designed within the two-semester timeline. This is the chance to work on something that interests us with hands-on experience. Don’t worry about competition as much as focusing on the project as it is.”

Extra note: Focus on the weight of the drone; it needs to be lightweight for the motors to carry it. Many students will overlook this concept and design a drone that is too heavy to fly.

Appendix B

Interview Questions for EDT

1. How much autonomy do you expect the drone to have versus how much can be driven manually?

“Within the context of the competition, the drone is supposed to be fully autonomous, however for safety and testing purposes we can manually take over and fly the drone.”

2. Is there a specific payload in mind?

“Currently, the payload is a roadside emergency flare beacon.”

3. How long should the drone be able to go on one charge?

“Normally it should be around 20-30mins.”

4. What kind of drone are we expected to design in terms of build? How will it carry the payload, release the payload, and what kind of controller is needed?

“We will be designing a quadcopter with an x-frame. We already have the FPV headset + camera purchased. We also already have the controller set up correctly. We will have a flight controller running ArduPilot for flight, and onboard an Nvidia Jetson Orin Nano computer for running object detection scripts.”

5. What are the size and weight constraints of the airdrop system?

“The lowest that the drone can hover is 50ft. Last year, we previously used a claw air drop system. A motor would be connected to a 10mm rod wrapped with 50ft of fishing line. The fishing line would be tied to the claw. The 3-finger claw design that would allow us to open and close the beacon (once lowered down 50ft). The claw had a separate micro-controller, motor, and battery. Each payload system is supposed to be less than 2lbs from last year's rules.”

Appendix C

Concept Map for our Design Alternatives

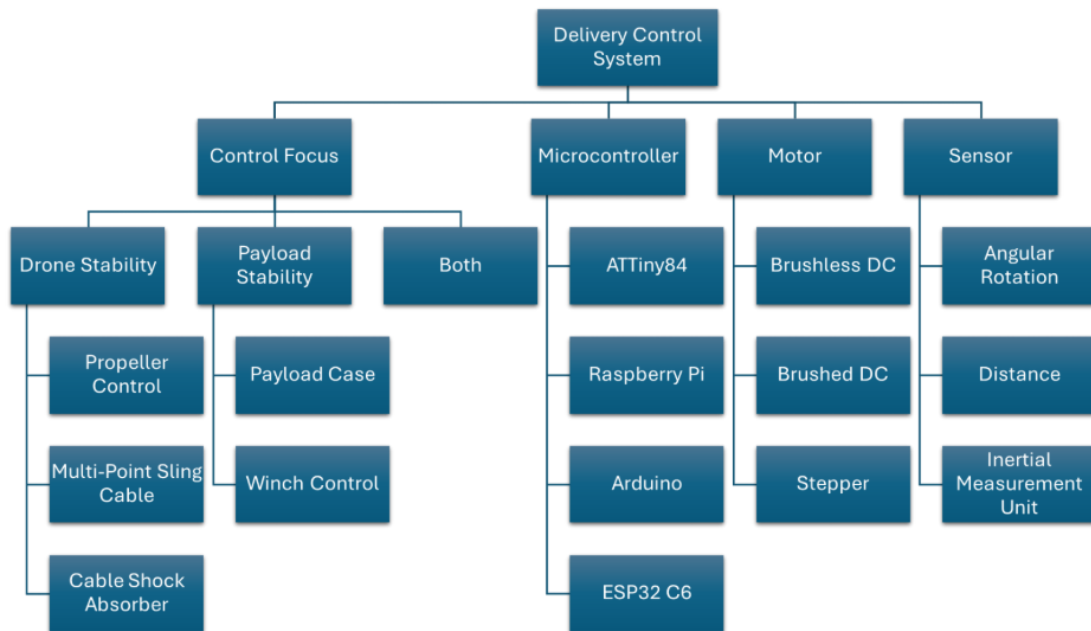


Figure C1: Concept map

Appendix D

Samuel's Pair-Wise Comparison

	Development Cost	Contribution to ECE Education	Design Complexity	Technical Knowledge	Time Allocation	Geometric Mean	Weights
Development Cost	1	9	3	3	5	3.32	0.48
Contribution to ECE Education	1/9	1	1/5	1/5	1/3	0.27	0.04
Design Complexity	1/3	5	1	1	3	1.38	0.20
Technical Knowledge	1/3	5	1	1	3	1.38	0.20
Time Allocation	1/5	3	1/3	1/3	1	0.58	0.08
Total							1.00

Appendix E

Jeremy's Pair-Wise Comparison

	Development Cost	Contribution to ECE Education	Design Complexity	Technical Knowledge	Time Allocation	Geometric Mean	Weights
Development Cost	1	1/7	1/5	1/5	1/9	0.23	0.03
Contribution to ECE Education	7	1	5	3	7	3.74	0.49
Design Complexity	5	1/5	1	3	9	1.93	0.25
Technical Knowledge	5	1/3	1/3	1	7	1.31	0.17
Time Allocation	9	1/7	1/9	1/7	1	0.46	0.06
Total							1.00

Appendix F

Stuart's Pair-Wise Comparison

	Development Cost	Contribution to ECE Education	Design Complexity	Technical Knowledge	Time Allocation	Geometric Mean	Weights
Development Cost	1	9	1	3	5	2.67	0.39
Contribution to ECE Education	1/9	1	1/5	1/3	1/3	0.30	0.04
Design Complexity	1	5	1	3	5	2.37	0.35
Technical Knowledge	1/3	3	1/3	1	1	0.80	0.12
Time Allocation	1/5	3	1/5	1	1	0.65	0.10
Total							1.00

Appendix G

Adrian's Pair-Wise Comparison

	Development Cost	Contribution to ECE Education	Design Complexity	Technical Knowledge	Time Allocation	Geometric Mean	Weights
Development Cost	1	1/5	1/3	1/3	1/5	0.34	0.06
Contribution to ECE Education	5	1	1/3	3	3	1.72	0.30
Design Complexity	3	3	1	1	1	1.55	0.27
Technical Knowledge	3	1/3	1	1	3	1.25	0.22
Time Allocation	5	1/3	1	1/3	1	0.89	0.15
Total							1.00

Appendix H

Kyle's Pair-Wise Comparison

	Development Cost	Contribution to ECE Education	Design Complexity	Technical Knowledge	Time Allocation	Geometric Mean	Weights
Development Cost	1	7	5	1/3	3	2.04	0.36
Contribution to ECE Education	1/7	1	3	5	1/3	0.93	0.16
Design Complexity	1/5	1/3	1	1/3	3	0.58	0.10
Technical Knowledge	3	1/5	3	1	5	1.55	0.27
Time Allocation	1/3	3	1/3	1/5	1	0.58	0.10
Total							1.00

Appendix I

John's Pair-Wise Comparison

	Development Cost	Contribution to ECE Education	Design Complexity	Technical Knowledge	Time Allocation	Geometric Mean	Weights
Development Cost	1	3	5	3	1	2.14	0.38
Contribution to ECE Education	1/3	1	1	5	3	1.38	0.24
Design Complexity	1/5	1	1	1	3	0.90	0.16
Technical Knowledge	1/3	1/5	1	1	3	0.72	0.13
Time Allocation	1	1/3	1/3	1/3	1	0.52	0.09
Total							1.00

Appendix J

Draft User Manual

I. Product Introduction

The payload case system is built with a 4-in-1 electronic speed controller, 4 motors with propellers, one Arduino, one inertial measurement unit, one altitude sensor, one ultrasonic sensor, and one servo. The control system is an algorithm that will use the inertial measurement units to help counteract the environmental forces acting on the drone as it conducts a delivery and keeps it safe during transit.

II. Getting Started

Here will provide step-by-step instructions for setting up and getting started.

a. Unpacking and Assembly

The payload case comes with all the electronic preinstalled. The MR30-M connectors for the motors and the XT60 connector for the battery will come disconnected. Before operational use, connect them.

b. Installation and Setup

This payload case is meant to work in tandem with a drone. The drone must have a winch mechanism for ideal operational use. There will be a mount where the drone's tether mechanism, such as a claw, needs to be hooked on.

c. Initialization Sequence

To activate the system, ensure that the LiPo battery is plugged in. You can tell the microcontroller is receiving power when the LED on the Arduino is on. From here, press the red button on the back of the payload case. This will begin the startup sequence for the payload case, indicated by each propeller turning on consecutively for a short period. After each propeller activates briefly, the system will begin its calibration state, which is indicated by a 10 second pause. During this time, ensure that you do not move the payload case and make sure that it is level, so that the gyroscope can take accurate data about the calibrated position. After calibrating, all propellers will turn on simultaneously for a short period, indicating that the system is now fully active.

d. Basic Operation

The operation of the control system is done entirely autonomously. While the drone is using its winch and claw system to either pick up or drop off the payload, the payload case will autonomously adjust the propellers if there are any forces affecting its orientation. This is all done as drone remains in a stable position as it completes the task. When the user wants to open/close the payload compartment to load/unload a package, they only need to wave their hand above an ultrasonic sensor.

III. Advanced Features

The control system can sense how high it is in the air and will deactivate the control system while the drone is in transit to location. This ensures the drone can fly without the payload case to affect its flight operation.

IV. Maintenance and Cleaning

Proper maintenance and cleaning of the product will maintain its performance and longevity. Do not let the payload case sit in direct sunlight for a long period of time. Do not store the payload case outside for a long period of time. After long continuous use, check the components to make sure everything is working properly. If dust builds after drone use, you can use a gentle brush to clean the dust off the components. You may need to replace sensors if they become faulty.

V. Technical Specifications

The system is designed to operate using one LiPo battery equipped with an XT60 connector and rated for a nominal voltage of 11.1V, with a minimum safe voltage of 9.0V. The payload case measures approximately 28 cm x 36 cm x 15 cm and has an estimated weight of 4.8 pounds. These specifications are provided to ensure a reliable operation of the winch system across a range of flight conditions and performance expectations.

VI. Safety Procedures

a. Electrical Handling

The payload case is powered through a 4-in-1 ESC, which can present a shock and become a damage hazard for the equipment if handled incorrectly. Always disconnect power before performing any work on the ESC, Arduino, sensors, or wiring, and wait at least 30 seconds to allow residual charge to dissipate. Never handle wiring or electronics while the propellers are installed, and the system is armed, as unexpected motor spin-up can occur, potentially causing severe injury.

b. LiPo Battery Safety

LiPo batteries must be used and handled carefully to prevent fire, swelling, or permanent damage to the system. Only use batteries that meet the specified voltage and connector requirements for this design. Do not operate the drone if any battery appears swollen, damaged, leaking, or excessively hot to the touch. Follow

EDT's approved charging and storage procedures at all times, and never leave batteries unattended while charging, even for short periods.

c. Mechanical Safety

Reliable operation depends on secure mechanical connections throughout the system. Before each flight, verify that all connectors to the ESC, IMU, altitude sensor, ultrasonic sensor, and servo are fully seated. Confirm that the payload case is firmly mounted and aligned to the drone as intended, as any shifting mass or loose hardware can alter the center of gravity and negatively affect flight stability. Ensure all four motors and propellers are securely fastened and free of obstruction before powering the system.

d. Operational Safety

Safe operation requires proper system configuration and good situational awareness during flight. The payload case control system deactivates automatically during transit based on altitude sensor readings, but operators should verify this behavior is functioning correctly before each mission. Maintain a clear safety perimeter around the drone during takeoff, hover, and payload delivery, keeping all personnel at a safe distance to avoid injury from the drone propellers, payload case propellers, moving payloads, or unexpected vehicle motion. Never place hands near the payload case propellers while the system is powered, including during package loading and unloading, even when triggering the ultrasonic sensor to open the compartment.

VII. Conclusion

This manual has outlined the essential steps to set up, operate, and maintain the payload case's control system and supporting hardware. Each component is essential to ensure stability and reliable performance. The autonomous control features allow the payload case to compensate forces during payload pickup and delivery, while the failsafe overrides provide flexibility when user intervention is needed. To keep the system functioning at peak performance, regular inspection and cleaning of hardware components is recommended.

Avoid exposing the drone to harsh environmental conditions and replace any sensors or power distribution components that are showing signs of failure or inconsistent output. By following these guidelines, users can ensure a smooth operation of extended lifespan hardware, as well as safe, dependable performance of the drone through its mission tasks.

Appendix K

Unit Testing: IMU

```
Power ON: Initializing system...
IMU connected, calibrating...
Calibration done!
Initial Roll: -0.82
Initial Pitch: -3.19
System Power ON
Roll: -0.79 | Pitch: -3.22
Roll: -0.79 | Pitch: -3.21
Roll: -0.78 | Pitch: -3.18
Roll: -0.77 | Pitch: -3.16
Roll: -0.74 | Pitch: -3.11
Roll: -0.72 | Pitch: -3.10
Roll: -0.73 | Pitch: -3.10
Roll: -0.71 | Pitch: -3.09
Roll: -0.71 | Pitch: -3.11
Roll: -0.68 | Pitch: -3.11
```

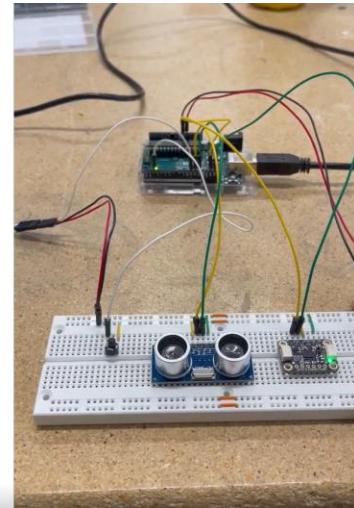


Figure K1: Setup for unit and integration testing with IMU and push button

Appendix L

Unit Testing: Ultrasonic Sensor and Servo

```
Distance (cm): 56
Distance (cm): 57
Distance (cm): 57
Distance (cm): 57
Distance (cm): 57
Distance (cm): 58
Distance (cm): 56
Distance (cm): 59
Distance (cm): 57
Distance (cm): 58
Distance (cm): 57
Distance (cm): 58
Distance (cm): 58
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Distance (cm): 66
Distance (cm): 56
Distance (cm): 55
Distance (cm): 55
Distance (cm): 55
```

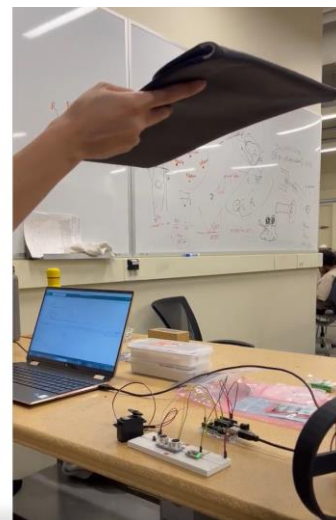


Figure L1: Setup for unit and integration testing with ultrasonic sensor and servo

Appendix M

Unit Testing: ESC and Motors

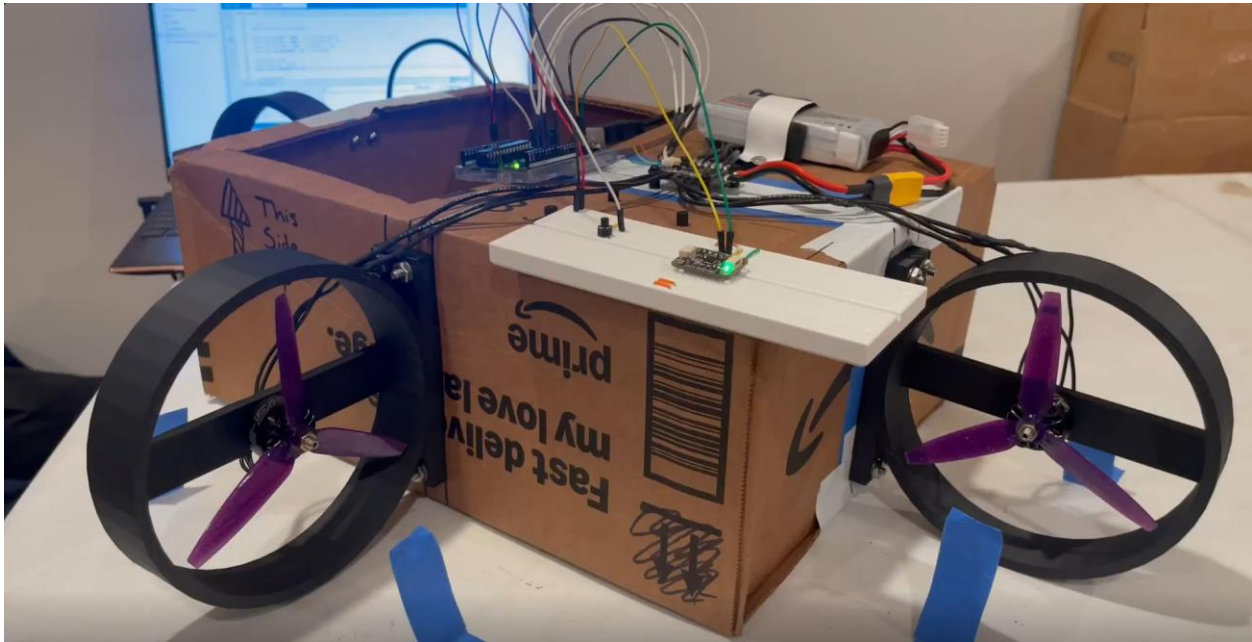


Figure M1: Setup for unit and integration testing with 4-in-1 ESC and motors

Appendix N

Final Prototype

Our final prototype configuration with updated case material and motor mounts can be found in Figure N1 and Figure N2. The hardware was placed on the inside of the case, attached to the top plate. Figure N1 is an image of the final prototype presented at the UIC Engineering Expo 2026. Figure N2 is an image of the Engineering Design Team's drone and our AirDrop Transport Companion.



Figure N1: AirDrop Transport Companion at UIC Engineering Expo 2026



Figure N2: EDT Drone Mounted on the AirDrop Transport Companion

Appendix O

Time Allocation

Table O1 is an estimated timeline for the team's task for the Spring 2026 semester. It includes tasks for the EXPO and the team member's name who will take the lead in ensuring the task is accomplished.

Table O1: Task Allocation and Timeline (Spring 2026 Semester)			
Week	Date	Task	Member Lead(s)
1	Jan 12	Confirm PDB schematic before buying parts	Sam
2	Jan 19	Buy parts for PDB and control system	Jeremy
3	Jan 26	Build PDB and test voltage ratings	Stuart
4	Feb 2	Implement PDB and hardware for the control system for the drone	Adrian
5	Feb 9	Expo student registration and project information submission (Project title and abstract) (due March 8 for student registration) (due April 8 for project information)	Kyle
6	Feb 16	Test/debug control algorithm ensuring it doesn't affect flight operation	Stuart
7	Feb 23	Test/debug control algorithm ensuring it can combat pendulum sway	Jeremy
8	Mar 2	Test/debug control algorithm ensuring it can take user's hand wave gesture input	Sam
9	Mar 9	Test/debug control algorithm ensuring fail-safes are in place and work	John
10	Mar 16	Finalize testing for control algorithm	Adrian
11	Mar 23	Create project document which is a one-page handout with summary of research, results, and costs (due April 15)	Kyle
12	Mar 30	Create the scientific poster (due April 15)	Stuart
13	April 6	Finalize project title, abstract, poster, and project documents for submission to the Expo student portal	John
14	April 13	Practice presentation for Expo	Jeremy

Appendix P

Obstacle and Error Log

Figure P1 and Figure P2 detail 6 different obstacles that our team ran into. The summary of each obstacle, the cause, the resolution, and the takeaway lessons are listed as well.

Entry #	Date	Phase	Obstacle Summary	Root Cause	Resolution	Lessons Learned	Status
1	1/21/2026	A	After doing research on pendulum motion, our team noticed that implementing a pendulum controls system for our drone would be quite difficult and might not be sufficient for our project.	The decision matrix picked that the pendulum system would be our best option for designing a payload delivery system, however, we are noticing some difficulty with implementing it well.	We decided we are going to create a payload system similar to that of the company Zipline, where they use a payload box with fans and a microcontroller that can control the accuracy of landing in a more mechanical fashion.	We need to make the decision matrix with better criteria and weighting so that our system fits our needs and constraints.	Resolved
2	2/4/2026	B	The parts we ordered are taking a while to arrive due to senior design department issues. This is delaying the prototyping process since we don't have any parts to use and test. This particular obstacle is unfortunately not in our control.	The root cause can be found with the time we ordered the parts and with the senior design department. A combination of these two causes created the delay.	We just have to wait longer for our parts, there isn't too much we can do about it right now.	In the future, we should consider ordering parts sooner than suggested in hopes to get the parts early so we can begin testing.	Resolved
3	3/4/2026	C	The software for the servo is different than we expected it to be. We expected to be able to control the servo's position manually using parameters, however, we are only able to edit the speed at which the servo turns and how long it turns for.	The root cause comes from the documentation that the seller gave upon purchasing our servo. We didn't recognize the possibility of moving our servo only through the use of turn rate and duration instead of position, like other servo's we have used in the past.	We are simply going to change our program to accommodate this servo feature so that it can still function as intended.	Next time, we should be very careful about buying a servo that allows for displacement parameters instead of rotation rate.	Resolved

Figure P1: Entry 1-3 of the Obstacle and Error Log

4	3/30/2026	D	Professor Alonso suggested taking measurements of altitude in order to determine when the payload case is operational. To do this we need an altitude sensor that can take these measurements. However, after purchasing the part and waiting, the part has still not arrived, meaning it might not be possible to implement it before expo.	The root cause comes from a lack of safety of the payload. Previously the motor mount did not have a fan guard to prevent users from accidentally touching the moving propellers. The altitude would make the payload safer for the user by disabling the fans when the payload has landed.	Right now, the solution is to create fan guards for the motor mount. However, the implementation of an altitude sensor is still desired due to the need to disable the system when it doesn't need to be active. We are planning to either use an ultrasonic sensor on the bottom of the payload or remove the altitude sensor system entirely.	Consider buying the altitude sensor beforehand and plan for the safety of the system better during the brainstorming and design phase.	Resolved
5	4/8/2026	D	We are currently not meeting a lot of engineering requirements because we are unable to produce a working PDB.	This problem comes from the fact that it took a while for the pcb to be delivered and assembled. It only got delivered recently, which isn't enough time to implement and test it before the expo.	The solution is to change the engineering requirements to not consider the PDB since most of our system is now hardware, utilizing a arduino uno, multiple sensors, and an esc. We still satisfy the hardware and software components of the project by removing the engineering requirements for the PDB.	Design and order the PCB a lot sooner than we did in order to ensure enough time to test and implement the system. This also means to plan a lot earlier in the semester and take action quicker.	Resolved
6	4/24/2026	Expo	For our judges results, we got near perfect scores for the demo, prototype design, and functionality. However, we were lacking in the presentation.	The root cause could be lack of preparation for presenting the project at expo, not recognizing who where judges and who were students, or effort put into presenting professionally at expo.	The resolution is more of a future solution. The goal is to take this expo, and use it to become a better presenter in the future.	Presenting is just as important as making the design, because it shows people that you care about the project and you worked hard on it. The perfect project can be scored last if the presenters don't do a good job.	Resolved

Figure P2: Entry 4-6 of the Obstacle and Error Log

Appendix Q

Individual Takeaways

Jeremy: It was a good opportunity to work with a team where our end goal was to build something that addresses real world issues. The whole process of documenting our research, testing, and revising our designs gave me insight into what I might need to do for future projects working with other engineers in the future.

Kyle: I got a great sense of working in a long-term team environment, and I feel like it gave me a great perspective on real-life work with group projects that we might all be working on someday. Especially with managing schedules and meeting deadlines.

Samuel: The research and planning stage is very important for the organization of a project. Having a thorough plan, correct calculations, and budget calculation will make the testing and development stage a lot smoother. If there is anything that this project has taught me, it is to make a solid plan with potential backup plans so that the design process is a lot less stressful.

Stuart: I am grateful for the opportunity to work alongside such a talented and dedicated group of people. Working through challenges together taught me the importance of communication, collaboration, and trusting the abilities of others. I am proud of the progress we made as a team, and I am even more proud to call these individuals my teammates because of the effort, dedication, and support they showed throughout the process.

John: This project gave me valuable hands-on experience bridging theoretical electrical engineering concepts with physical hardware integration. Working together on the payload system and troubleshooting components as a team has been great preparation for future engineering projects, where practical problem solving and system reliability are essential.

Adrian: This project was a great opportunity to work as a team to utilize knowledge from prior ECE courses in an application to address a real-world issue. The process of research, documentation, organization, prototyping, and presentation was an experience that helped me prepare for future research and design projects in a professional environment.